

DBMS – Exam Preparation Notes

1. Database Management System

- A DBMS is software used to create, store, organize, retrieve, and manage data in databases.
- Major DBMS functions include data definition, data manipulation, transaction management, security, backup, and recovery.

2. Database Models

- The relational model organizes data into tables consisting of rows and columns.
- Other common models include hierarchical, network, object-oriented, and document-oriented models.

3. Keys

- A primary key uniquely identifies a record in a relation.
- A foreign key references a key in another relation and helps establish relationships between tables.
- Candidate keys are attributes or attribute sets that can uniquely identify tuples.

4. Normalization

- Normalization organizes relations to reduce redundancy and update anomalies.
- First Normal Form requires atomic attribute values.
- Second Normal Form requires 1NF and removal of partial dependency on a composite candidate key.
- Third Normal Form requires 2NF and removal of transitive dependency.

5. SQL

- DDL commands define database structures, such as CREATE, ALTER, and DROP.
- DML commands manipulate data, such as SELECT, INSERT, UPDATE, and DELETE.
- A JOIN combines related rows from multiple tables.

6. Transactions

- A transaction is a logical unit of database work.
- ACID properties are Atomicity, Consistency, Isolation, and Durability.
- Concurrency control helps maintain correctness when multiple transactions execute simultaneously.